

PeerPulse AI

Product Requirements Document (PRD)

Peer-Relative Profiling of District Blood-Bank Service Gaps in India

Version: 1.0 **Status:** Development

A decision-support web application that converts ASAR/NACO district blood-bank data into explainable peer-relative profiles.

1. Product Overview

PeerPulse AI analyzes district-level blood-bank data and identifies service gaps by comparing each district with statistically similar districts. The core question is: **How is this district performing compared with similar districts?**

2. Problem Statement

District blood-bank performance varies across India. Simple national averages can hide important differences between districts. PeerPulse AI creates peer groups of comparable districts and measures each district's performance relative to those peers.

3. Product Goal

Build a web application that transforms ASAR/NACO blood-bank data into an understandable district-level peer-relative profiling system. Users should be able to select a district, view its profile, inspect its peers, understand its gaps, compare districts, and review the methodology.

4. Target Users

Researchers/Students: analyze district-level patterns and generate evidence. **Policy/Public-health Analysts:** identify relative underperformance. **Program Managers:** understand service dimensions. **Academic Evaluators:** review methodology and validation.

5. Data Sources

The initial system uses ASAR/NACO 2016 blood-banking datasets. The district dataset contains 616 original rows, with 615 usable districts after preprocessing, across 35 states/UTs and 39 variables. The state dataset contains 35 states/UTs and 304 variables and is retained for future contextual analysis. The initial peer-relative model uses the district dataset.

6. Core Features

District Search: search/select a district. **District Profile:** show district, state, indicators, peer group, gaps and primary gap. **Peer Group:** show the 20 comparable districts. **District Comparison:** compare multiple districts. **Dashboard:** show district/state totals, gap distributions, indicator distributions and peer-relative rankings.

7. Analytical Model

The frozen primary model uses three features: **bb_availability**, **district_annual_per100**, and **percentage_voluntary_district**. Features are standardized with StandardScaler. KNN uses **K=20** and **Euclidean distance**. Mahalanobis distance is retained as a sensitivity check.

8. Gap Detection

For each district, performance is compared with its peer-group performance across **Availability**, **Collection**, and **Voluntary Donation**. The strongest relative shortfall is identified as the primary gap. The primary gap is an analytical signal, not proof of causality.

9. Validation and Robustness

Current implementation results:

Validation	Result
K=10 vs K=20 agreement	74.80%
K=30 vs K=20 agreement	86.34%

Resampling stability — mean	98.16%
Resampling stability — minimum	95.01%
Resampling stability — maximum	99.89%
Euclidean vs Mahalanobis agreement	80.00%

Decision: retain Euclidean + $K=20$ as the primary configuration and use Mahalanobis as a sensitivity check. K sensitivity shows that individual labels can change, while the resampling analysis supports strong stability of the overall gap-type distribution.

10. Backend Architecture

Raw ASAR/NACO data → Data Loader → Preprocessing → Standardization → KNN Peer Matching → Gap Detection → Validation/Stability → Processed Results → Backend API → Frontend.

11. Backend API

GET /api/districts — district list. **GET /api/districts/{id}** — district profile. **GET /api/districts/{id}/peers** — 20 peers. **GET /api/districts/{id}/gaps** — peer-relative gaps. **GET /api/summary** — dashboard statistics. **GET /api/validation** — validation results.

12. Frontend Requirements

Recommended pages: Dashboard (/), District Explorer (/districts), District Profile (/district/:id), District Comparison (/compare), and Methodology & Validation (/methodology). The interface should be searchable, simple, research-oriented and explainable.

13. Technology Stack

Data/ML: Python, Pandas, NumPy, SciPy, Scikit-learn, Statsmodels. **Visualization:** Matplotlib, Seaborn. **Backend:** FastAPI. **Frontend:** Next.js, React, TypeScript. **Prototype database:** SQLite; PostgreSQL later if needed.

14. Processing Pipeline

Raw Excel → Data Audit → Missing-value handling → Feature Selection → Standardization → KNN Peer Matching → Peer Mean Calculation → Gap Calculation → Primary Gap Detection → Validation → Processed Results → API → Frontend.

15. Non-Functional Requirements

Performance: district profiles should load quickly after analytical results are precomputed. **Reproducibility:** identical data and frozen parameters should reproduce results. **Explainability:** every result should be traceable to district data, peers, peer statistics and gap calculations. **Transparency:** methodology and limitations must be visible.

16. Limitations

The dataset is from 2016 and is cross-sectional. Analysis is district-level. Peer-relative gaps do not establish causality. A gap does not automatically imply a specific intervention. $K=20$ is a selected configuration, not a universally optimal value. Distance-metric sensitivity exists, and missing data required preprocessing.

17. Out of Scope — Version 1

Real-time blood-bank inventory, donor matching, hospital emergency management, blood-demand forecasting, patient-level prediction, medical diagnosis, automated policy recommendations, causal inference, and live NACO integration.

18. Future Versions

V2: state-level comparison, additional contextual features, interactive peer visualization and PDF reports. **V3:** time-series data, shortage forecasting, real-time inventory and alerts. **V4:** advanced ML, explainable AI, policy simulation and federated/multi-institution learning if suitable data becomes available.

19. MVP Definition

A working system that takes the ASAR/NACO district dataset, creates 20 comparable peers for each district, calculates peer-relative gaps across the three frozen dimensions, identifies the primary gap, validates the methodology, and exposes results through a usable web interface.

20. MVP Completion Checklist

Completed: dataset loading; data audit; missing-data handling; three-feature freeze; standardization; KNN peer engine; K=20 selection; gap calculation; primary-gap detection; K sensitivity; resampling stability; Mahalanobis sensitivity. **Next:** final validation notebook; final results tables; backend API; frontend dashboard; district profile; peer visualization; testing; deployment.

21. Development Roadmap

04_validation.ipynb → **05_results.ipynb** → **Freeze ML outputs** → **FastAPI backend** → **Next.js frontend** → **Connect frontend/backend** → **Testing** → **Deployment**. The data/ML work is the analytical intelligence phase; the frontend and API follow after the analytical pipeline is frozen.